# Universidad Carlos III de Madrid

Bachelor in Aerospace Engineering


Mechanics Applied to Aerospace Engineering

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## Particle on Oscillating Loop

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Laboratory Session 2


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# 1 Introduction

In this laboratory report, we will investigate the dynamics of a heavy point particle $P$ of mass $m$ that slides without friction on a circular wire of centre $C$ and radius $R$. The wire is on a vertical plane $Oxy$ that rotates about the vertical axis $Oy$ with a known angular velocity $\boldsymbol{\Omega}(t) = A\omega \sin(\omega t)\,\mathbf{j}$, and the position of $P$ on the wire is given by the angle $\varphi$ between $\overrightarrow{CP}$ and $Ox$ (Fig. 1).

The main difficulty is that the frame where the motion can be easily described, the one attached to the wire, is not inertial and does not rotate at a constant rate, so the centrifugal, Euler and Coriolis forces all act on the particle. Our objective is to derive the equation of motion and the reaction of the wire, to integrate it in Matlab with `ode45` and explain the forces and the energy we obtain, to find a condition that tells if the particle oscillates or rotates on a fixed loop and check it with simulations, and to see how far we can trust a linearized equation around the lowest point of the wire, where resonance appears.

In Section 2 we derive the analytical model and in Section 3 we explain how we have implemented it in Matlab. In Section 4 we present the results and discuss each of them next to its figure, and Section 5 are the conclusions .

# 2 Analytical Model

## 2.1 Reference Frames and Degrees of Freedom

$Oxyz$ rotates with the plane of the wire: $\mathbf{i}$ is horizontal along the diameter through $C$, $\mathbf{j}$ is vertical and upward (the rotation axis) and $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ is normal to the plane, pointing out of the paper in Fig. 1. $Ox_1y_1z_1$ is a stationary frame, assumed inertial, that coincides with $Oxyz$ at $t = 0$, with $\mathbf{j}_1 = \mathbf{j}$. The axis $Oy$ is tangent to the wire at $O$, so $\overrightarrow{OC} = R\,\mathbf{i}$.

In space, $P$ needs three coordinates. The wire removes two of them, since $P$ has to stay in the plane $Oxy$ and at a distance $R$ from $C$, which means that the problem only has $3 - 2 = 1$ configuration degree of freedom ($\varphi$). The position of $P$ seen from the stationary frame depends on $\varphi$ and also explicitly on $t$.

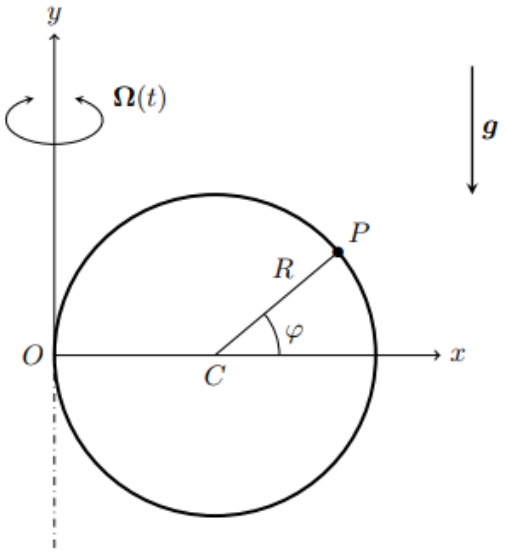



**Figure 1:** Geometry of the problem and reference frames.

The angle $\theta$ turned by the plane satisfies $\dot{\theta} = \Omega$ with $\theta(0) = 0$, where $\Omega = A\omega \sin \omega t$. Integrating with the change $u = \omega\tau$, $du = \omega\, d\tau$:

$$\theta(t) = \int_0^t A\omega \sin(\omega\tau)\, d\tau = A \int_0^{\omega t} \sin u\, du = A\big[-\cos u\big]_0^{\omega t} = A\,(1 - \cos \omega t), \tag{2.1}$$

so $\theta$ oscillates between $\theta = 0$ and $\theta = 2A$ with period $T = 2\pi/\omega$. Using this angle $\theta$ to get the rotating reference frame $Oxyz$, we obtain these unit vectors in function of the stationary frame's unit vectors:

$$\mathbf{i} = \cos\theta\,\mathbf{i}_1 - \sin\theta\,\mathbf{k}_1, \quad \mathbf{j} = \mathbf{j}_1, \quad \mathbf{k} = \sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{k}_1, \tag{2.2}$$

### 2.2 Position, Velocity and Acceleration of $P$

The side adjacent to $\varphi$ is $R\cos\varphi$ along $\mathbf{i}$ and the opposite one is $R\sin\varphi$ along $\mathbf{j}$. Then, with $x = R(1+\cos\varphi)$ the distance from $P$ to the rotation axis,

$$\mathbf{r} = \overrightarrow{OC} + \overrightarrow{CP} = R(1+\cos\varphi)\,\mathbf{i} + R\sin\varphi\,\mathbf{j} = x\,\mathbf{i} + R\sin\varphi\,\mathbf{j}. \tag{2.3}$$

Using the auxiliary polar reference frame defined as:

$$\mathbf{e}_r = \cos\varphi\,\mathbf{i} + \sin\varphi\,\mathbf{j}, \quad \mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i} + \cos\varphi\,\mathbf{j}, \tag{2.4}$$

With the time derivatives of the unit vectors being:

$$\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi, \quad \dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r, \tag{2.5}$$

Also,

$$\mathbf{i} = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi, \quad \mathbf{j} = \sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi, \tag{2.6}$$

Differentiating in the $Oxyz$ reference frame applying the chain rule, we get the velocity and acceleration:

$$\mathbf{v}_{rel} = -R\dot\varphi\sin\varphi\,\mathbf{i} + R\dot\varphi\cos\varphi\,\mathbf{j} = R\dot\varphi\,\mathbf{e}_\varphi, \tag{2.7}$$

$$\mathbf{a}_{rel} = -R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi)\,\mathbf{i} + R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)\,\mathbf{j} = R\ddot\varphi\,\mathbf{e}_\varphi - R\dot\varphi^2\,\mathbf{e}_r. \tag{2.8}$$

### 2.3 Forces on the Particle

Two real forces act on $P$: its weight and the reaction of the wire. The wire is frictionless, so the reaction has no component along the tangent $\mathbf{e}_\varphi$:

$$m\mathbf{g} = -mg\,\mathbf{j} = -mg\,(\sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi), \qquad \mathbf{N} = N_1\,\mathbf{e}_r + N_2\,\mathbf{k}. \tag{2.9}$$

Since $Oxyz$ is not inertial, we have to add the inertia forces. With $\mathbf{j}\times\mathbf{i} = -\mathbf{k}$, $\mathbf{j}\times\mathbf{j} = \mathbf{0}$ and $\mathbf{j}\times\mathbf{k} = \mathbf{i}$ we get $\boldsymbol{\Omega}\times\mathbf{r} = -\Omega x\,\mathbf{k}$, $\dot{\boldsymbol{\Omega}}\times\mathbf{r} = -\dot\Omega x\,\mathbf{k}$, $\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}) = -\Omega^2 x\,\mathbf{i}$ and $\boldsymbol{\Omega}\times\mathbf{v}_{rel} = \Omega R\dot\varphi\sin\varphi\,\mathbf{k}$, and we also use $\mathbf{i} = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi$:

$$\mathbf{F}_{cent} = -m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}) = m\Omega^2 x\,\mathbf{i} = mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\,[\cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi], \tag{2.10}$$

$$\mathbf{F}_{Euler} = -m\dot{\boldsymbol{\Omega}}\times\mathbf{r} = m\dot\Omega x\,\mathbf{k} = mA\omega^2 R\cos(\omega t)(1+\cos\varphi)\,\mathbf{k}, \tag{2.11}$$

$$\mathbf{F}_{Cor} = -2m\boldsymbol{\Omega}\times\mathbf{v}_{rel} = -2mA\omega R\dot\varphi\sin(\omega t)\sin\varphi\,\mathbf{k}. \tag{2.12}$$

### 2.4 Equation of Motion and Reaction Forces

In $Oxyz$, $m\mathbf{a}_{rel} = m\mathbf{g} + \mathbf{N} + \mathbf{F}_{cent} + \mathbf{F}_{Euler} + \mathbf{F}_{Cor}$. The direction $\mathbf{e}_\varphi$ contains no reaction, and the Euler and Coriolis forces are along $\mathbf{k}$, so projecting on $\mathbf{e}_\varphi$ gives $mR\ddot\varphi = -mg\cos\varphi - m\Omega^2 x\sin\varphi$, that is

$$\boxed{\ddot\varphi = -\frac{g}{R}\cos\varphi - A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi} \tag{2.13}$$

The other two projections give the reactions. On $\mathbf{e}_r$: $-mR\dot\varphi^2 = N_1 - mg\sin\varphi + m\Omega^2 x\cos\varphi$, and on $\mathbf{k}$: $0 = N_2 + m\dot\Omega x - 2m\Omega R\dot\varphi\sin\varphi$:

$$N_1 = -mR\dot\varphi^2 + mg\sin\varphi - mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\cos\varphi, \tag{2.14}$$

$$N_2 = mA\omega R\big[2\dot\varphi\sin(\omega t)\sin\varphi - \omega\cos(\omega t)(1+\cos\varphi)\big]. \tag{2.15}$$

### 2.5 Mechanical Energy and Work of the Wire

The absolute velocity is $\mathbf{v} = \dot{\mathbf{r}} + \boldsymbol{\Omega}\times\mathbf{r} = \mathbf{v}_{rel} + \boldsymbol{\Omega}\times\mathbf{r} = R\dot\varphi\,\mathbf{e}_\varphi - \Omega x\,\mathbf{k}$. Since $\mathbf{e}_\varphi \perp \mathbf{k}$, $|\mathbf{v}|^2 = R^2\dot\varphi^2 + \Omega^2 x^2$, so the kinetic energy is $E_k = \frac{1}{2}m|\mathbf{v}|^2$, and taking the height of $O$ as the reference for the potential energy,

$$E_k = \tfrac{1}{2}m\big(R^2\dot\varphi^2 + \Omega^2 x^2\big), \qquad E_p = mgR\sin\varphi, \qquad E = E_k + E_p. \tag{2.16}$$

By the work–energy theorem $dE_k/dt = (m\mathbf{g} + \mathbf{N})\cdot\mathbf{v}$, because in the stationary frame they are the only forces acting on $P$, and the weight is conservative, $m\mathbf{g}\cdot\mathbf{v} = -dE_p/dt$. Only the power of the reaction is left, and since $\mathbf{e}_r \perp \mathbf{e}_\varphi$ and $\mathbf{e}_r \perp \mathbf{k}$,

$$\frac{dE}{dt} = \mathbf{N}\cdot\mathbf{v} = (N_1\mathbf{e}_r + N_2\mathbf{k})\cdot(R\dot\varphi\,\mathbf{e}_\varphi - \Omega x\,\mathbf{k}) = -\Omega x N_2. \tag{2.17}$$

On a fixed wire the reaction is perpendicular to the velocity. Here the wire moves with the plane, so $N_2$ does work and $E$ is conserved if and only if $A = 0$.

### 2.6 Fixed Loop: Criterion for the Motion

When the wire is fixed ($A = 0$), the plane does not rotate ($\Omega = 0$), so the reaction $N_2$ does no work and the mechanical energy from Eq. (2.16) is strictly conserved ($dE/dt = 0$):

$$\frac{1}{2}\dot{\varphi}^2 + \frac{g}{R}\sin\varphi = \frac{1}{2}\dot{\varphi}_0^2 + \frac{g}{R}\sin\varphi_0 = \frac{E_0}{mR^2}. \tag{2.18}$$

The potential energy has its maximum, $mgR$, at the top of the loop ($\varphi = \pi/2$). If $E_0 > mgR$, $P$ reaches the top with $\dot{\varphi} \neq 0$, so $\dot{\varphi}$ never changes sign and $\varphi$ grows (or decreases) steadily. If $E_0 < mgR$, $P$ stops before the top and oscillates. Imposing $E_0 > mgR$ in Eq. (2.18), $P$ rotates if

$$|\dot{\varphi}_0| > \dot{\varphi}_c(\varphi_0) = \sqrt{\frac{2g}{R}(1 - \sin\varphi_0)}, \tag{2.19}$$

with $\dot{\varphi}_0 > 0$ for a growing $\varphi$. Otherwise, $\dot{\varphi} = 0$ in Eq. (2.18) gives the turning points, $\varphi_{\max} = \arcsin(\sin\varphi_0 + R\dot{\varphi}_0^2/(2g))$ and $\varphi_{\min} = -\pi - \varphi_{\max}$. The limit $E_0 = mgR$ is the separatrix $\dot{\varphi} = \pm\sqrt{(2g/R)(1 - \sin\varphi)}$ shown in Fig. 6.

### 2.7 Linearized Equation

With $\varphi = \delta - \pi/2$, $\cos\varphi = \sin\delta$, $\sin\varphi = -\cos\delta$ and $\ddot{\varphi} = \ddot{\delta}$, so Eq. (2.13) becomes the exact $\ddot{\delta} = -(g/R)\sin\delta + \Omega^2(1 + \sin\delta)\cos\delta$. For $\delta \ll 1$, $\sin\delta \approx \delta$ and $\cos\delta \approx 1$, which gives $\ddot{\delta} = -(g/R)\delta + \Omega^2(1 + \delta)$. The hypothesis $A^2 \ll g/(R\omega^2)$ means $\Omega^2 \leq A^2\omega^2 \ll g/R$, so $\Omega^2\delta$ is dropped against $(g/R)\delta$. Using $\sin^2\omega t = \frac{1}{2}(1 - \cos 2\omega t)$:

$$\ddot{\delta} + \frac{g}{R}\delta - \Omega^2 = 0 \quad\Longleftrightarrow\quad \ddot{\delta} + \omega_n^2\delta = F_0\,(1 - \cos 2\omega t), \qquad \omega_n = \sqrt{g/R}, \quad F_0 = \tfrac{1}{2}A^2\omega^2. \tag{2.20}$$

This is an undamped harmonic oscillator of natural frequency $\omega_n$ that is forced at the frequency $2\omega$ by the centrifugal force, which peaks twice per period of the plane. The guidelines give no initial conditions, so we take $P$ at rest at the bottom, $\delta(0) = \dot{\delta}(0) = 0$. When the forcing frequency is equal to the natural one, $2\omega = \omega_n$, that is $\omega = \frac{1}{2}\sqrt{g/R}$, we have resonance: every push of the centrifugal force arrives in phase with the natural oscillation, so the amplitude of $\delta$ grows with time and $\delta \ll 1$ stops being true.

## 3 Numerical Implementation in Matlab

Equation (2.13) has to be written for the state vector $X = [\varphi, \dot{\varphi}]^T = [X_1, X_2]^T$ because `ode45` only integrates first-order systems:

$$\frac{d}{dt}\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = f(t, X) = \begin{pmatrix} X_2 \\ -\dfrac{g}{R}\cos X_1 - A^2\omega^2\sin^2(\omega t)\,(1 + \cos X_1)\sin X_1 \end{pmatrix}. \tag{3.1}$$

This is the function `diffeq(t,X,g,omega,R,A)` of `main.m`, and Eq. (2.20) is `diffeqlin`, with $X = [\delta, \dot{\delta}]^T$. The options are `RelTol` $= 10^{-8}$, `AbsTol` $= 10^{-10}$ (the default $10^{-6}$ is not small against $\delta \sim 10^{-3}$) and `MaxStep` $= 0.01$ s, so every swing has many points. We compute the forces and energies afterwards from $\varphi$ and $\dot{\varphi}$ with Eqs. (2.10)–(2.15) and (2.16). The position in the stationary frame follows from Eqs. (2.1)–(2.3): $x_1 = x\cos\theta$, $y_1 = R\sin\varphi$, $z_1 = -x\sin\theta$. A fixed-loop case counts as rotating if $\dot{\varphi}$ never changes sign and as oscillating otherwise, and the periods come from the times of the extrema of $\varphi$.

In all cases $g = 9.81\,\mathrm{m/s^2}$ and $m = 1\,\mathrm{kg}$. The oscillating loop uses $A = \pi$, $R = 1\,\mathrm{m}$, $\omega = 0.1\,\mathrm{rad/s}$, $\varphi(0) = \dot{\varphi}(0) = 0$ and $t \in [0, 3T]$ with $T = 62.83\,\mathrm{s}$. The fixed loop uses $A = 0$ and $R = 1\,\mathrm{m}$, and the linearized study $A = 0.1$, $R = 0.01\,\mathrm{m}$ and $\omega_n = 31.32\,\mathrm{rad/s}$. The guidelines do not fix the duration of these two studies, so we take $t \in [0, 10]$ s, which is longer than any period of the fixed loop.

# 4 Results and Discussion

## 4.1 Angle and Reaction Forces

Figure 2 shows $\varphi$, $N_1$ and $N_2$ for $t \in [0, 3T]$, with $T = 62.83\,\text{s}$. There are too many swings in that interval to tell them apart, so the right panels zoom into a window of 10 s around $t = T/4$. At $t = 0$ the particle is at rest at the end of the horizontal diameter, so it falls and swings around the bottom of the wire, between $\varphi \approx 0$ and $\varphi \approx -\pi$ ($\varphi \in [-3.14, 0.01]$ rad), passing through $O$ at the left end. The motion is almost the one of a simple pendulum, and its amplitude does not change during the three periods of the plane, because $\omega = 0.1\,\text{rad/s}$ is small: the plane turns slowly and the centrifugal term of Eq. (2.13) is very small compared with $g/R$. For the same reason the period of the swing, 2.37 s, is the one of a pendulum with a large amplitude (90° to each side of the bottom), and it is longer than the period of small oscillations, $2\pi\sqrt{R/g} = 2.01\,\text{s}$, because a pendulum is slower when the amplitude is large.

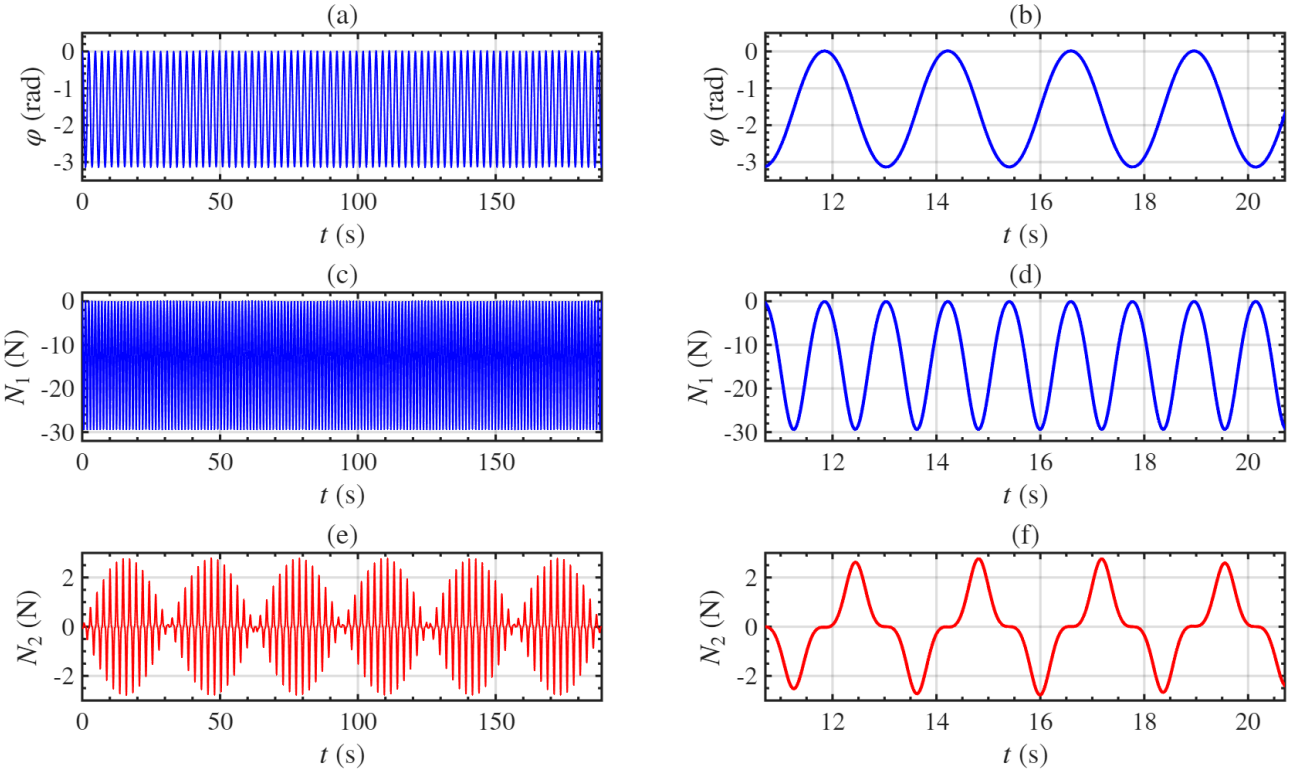



**Figure 2:** Oscillating loop. Left: $t \in [0, 3T]$. Right: window around $t = T/4$. (a, b) Angle; (c, d) reaction along $\mathbf{e}_r$; (e, f) reaction along $\mathbf{k}$.

$N_1$ is always negative, between $-29.43\,\text{N}$ and $0.00\,\text{N}$, so the wire pulls $P$ towards the centre $C$. It is zero at the turning points, where $P$ stops, and it has its minimum at the bottom, where the speed is largest ($\dot\varphi = 4.43\,\text{rad/s}$). As the rotation is weak, Eq. (2.18) is approximately valid and gives $\dot\varphi^2 \approx 2g/R$ at the bottom, so Eq. (2.14) gives $N_1 \approx -2mg - mg = -3mg$: the wire balances the weight and also pulls $P$ towards $C$ to make it turn. The particle crosses the bottom twice per swing, so $N_1$ oscillates at twice the frequency of $\varphi$ (Fig. 2d). $N_2$ is smaller, with a maximum of 2.78 N, and its sign changes from one crossing of the bottom to the next because $P$ goes through it in opposite directions (Fig. 2f). Its amplitude changes with a period $T/2$ and it is small when the plane stops and turns back ($\Omega \approx 0$). We explain where $N_2$ comes from in the next section.

## 4.2 Inertia Forces

Figure 3 shows the three inertia forces. Each has its own axis because their sizes are different, as the guidelines recommend. The centrifugal force (a) always points away from the rotation axis and its maximum is 0.20 N, reached when the plane turns fastest and $P$ is far from the axis. The Euler force (b) is proportional to $\dot\Omega$, so it follows $\cos\omega t$, and its maximum (0.06 N) is when the plane stops and changes its direction, which is when the centrifugal and Coriolis forces vanish. The Coriolis force (c)

is the largest, up to 2.78 N, because it depends on the velocity of $P$ along the wire, $\dot{\varphi}$, which is large while the plane turns slowly. It vanishes when the plane stops and at the turning points, where $\dot{\varphi} = 0$.

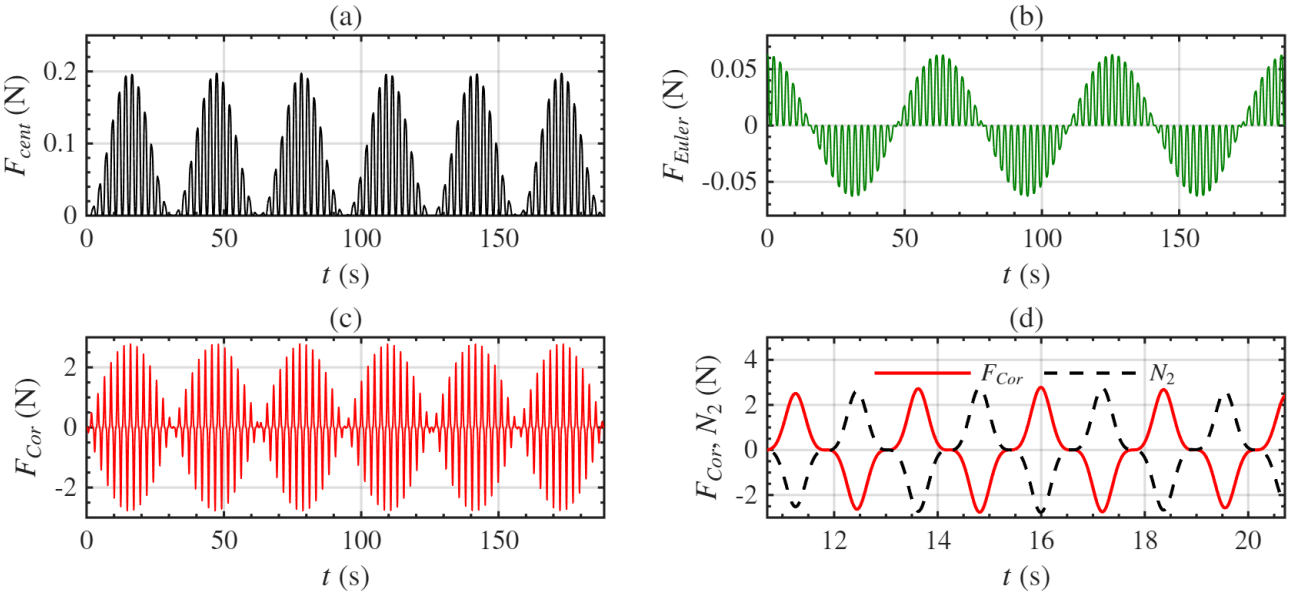



**Figure 3:** Inertia forces of the oscillating loop. (a) Centrifugal (along **i**), (b) Euler and (c) Coriolis (along **k**) for $t \in [0, 3T]$. (d) Coriolis force and reaction $N_2$ around $t = T/4$.

Only the centrifugal force has a component along $\mathbf{e}_\varphi$ and enters the equation of motion, Eq. (2.13), which is why $\varphi$ moves almost as in a pendulum. The Euler and Coriolis forces are along **k**, normal to the plane of the wire, so they do not move $P$ along the wire. They are balanced by $N_2$ (Eq. (2.15)), and this is why $N_2$ in Fig. 3d is almost the opposite of $F_{Cor}$, the Euler force being small.

### 4.3 Energy

In Fig. 4(a, b), $E_k$ and $E_p$ are exchanged in each swing: $E_k$ goes from zero at the turning points to its maximum, 9.83 J, at the bottom, where $E_p$ is minimum (−9.81 J), as in a pendulum. The total energy is not constant (Fig. 4c, d): $E - E_0$ varies between −0.08 J and 0.31 J. The reason is Eq. (2.17). The wire moves with the plane, so $N_2$ does work: when $P$ moves away from the rotation axis the wire gives energy to $P$, and when $P$ goes back to the axis the wire takes it back, so $E$ is high when $P$ is far from the axis and low when it is near (Fig. 4d). When the plane stops, $\Omega = 0$ and $E$ is again almost its initial value: $E - E_0$ is 0.001 J, 0.000 J and 0.001 J at $T$, $2T$ and $3T$. This is why the amplitude of $\varphi$ does not change in Fig. 2a.

### 4.4 Trajectory in the Stationary Frame

Figure 5 shows the trajectory of $P$ seen from the stationary frame. $P$ always stays on the surface swept by the wire when it turns about the vertical axis, which is a torus without a hole because the wire touches the axis at $O$. Each swing draws one petal, from the axis ($x = 0$, $\varphi = -\pi$) to $x = 2R$ ($\varphi = 0$) and back. The petals are spread around the axis because the plane turns from $\theta = 0$ to $\theta = 2\pi$ at $t = T/2$ and back to 0 at $t = T$ (Eq. (2.1)). Near $\theta = 0$ and $\theta = 2\pi$ the plane stops and reverses its rotation, so many petals accumulate in the same direction ($+x_1$ in Fig. 5b), while near $\theta = \pi$ the plane turns fastest ($\Omega$ is maximum) and the petals are more separated. The three periods of the plane draw almost the same figure, with slightly displaced petals because $T$ is not a multiple of the swing period. The video `trajectory_video.avi`, generated by `main.m`, shows the first period.

### 4.5 Fixed Loop: Phase Portrait and Validation of the Criterion

Figure 6 shows $\dot{\varphi}$ as a function of $\varphi$ for the nine initial conditions, together with the separatrix, which separates the two kinds of motion. Since $\varphi$ is plotted modulo $2\pi$, both ends of the horizontal axis are the top of the loop. Table 1 compares the criterion of Eq. (2.19) with the simulation (we consider that $P$ rotates if $\dot{\varphi}$ never changes sign), and they agree in all nine cases.

The closed curves, around the stable equilibrium $(-\pi/2, 0)$, are oscillations. The open curves are the

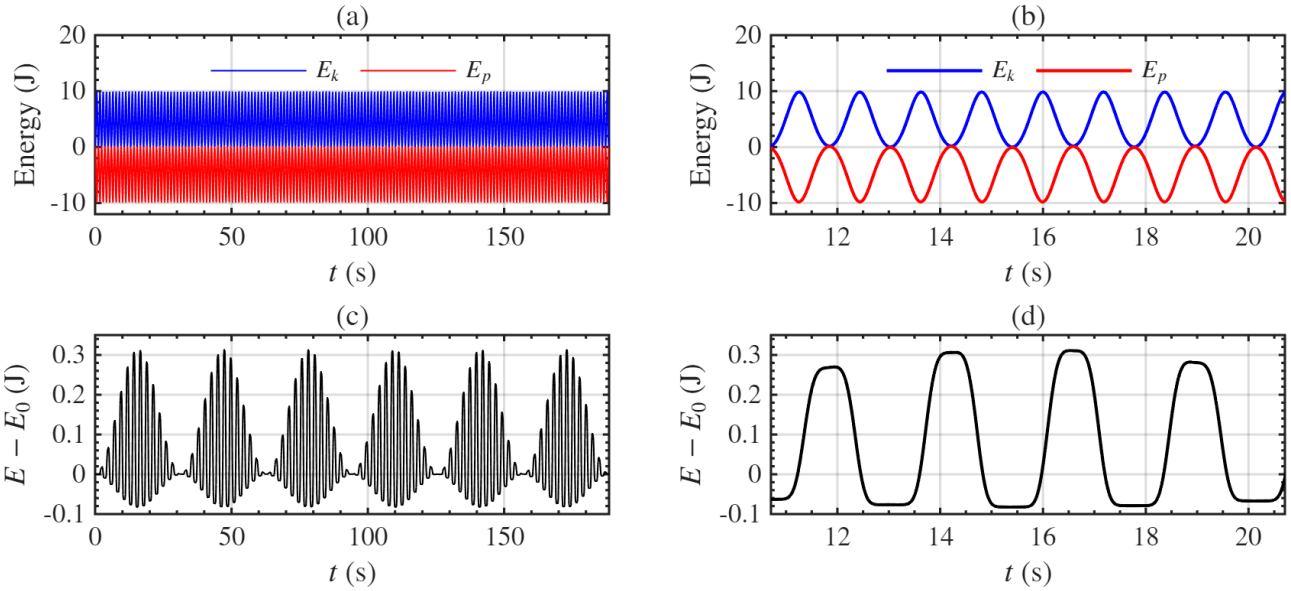



**Figure 4:** Energy of the oscillating loop. (a, b) Kinetic and gravitational potential energy; (c, d) change of the mechanical energy. Left: $t \in [0, 3T]$. Right: window around $t = T/4$.

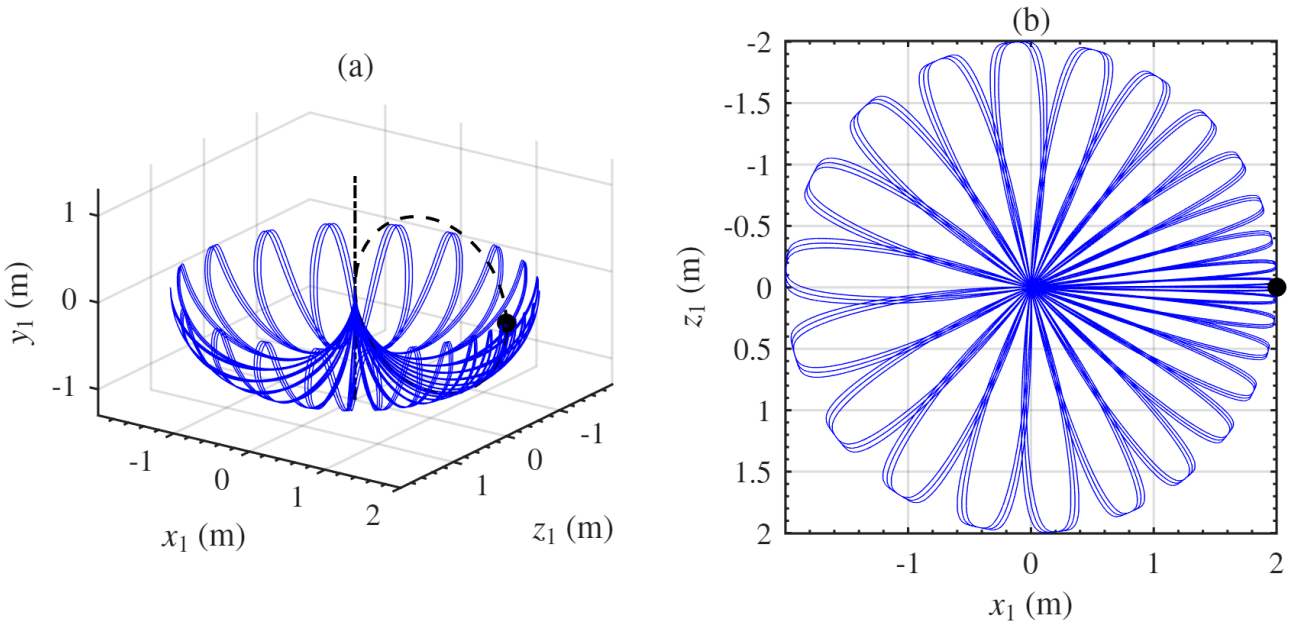



**Figure 5:** Trajectory of $P$ in $Ox_1y_1z_1$ for $t \in [0, 3T]$, with the start marked by a dot. (a) Three-dimensional view with the wire at $t = 0$ (dashed) and the rotation axis (dash-dot). (b) View from above.

**Table 1:** Criterion of Eq. (2.19) against the simulation of the fixed loop.

| $\varphi_0$ (rad) | $\dot{\varphi}_0$ (rad/s) | $\dot{\varphi}_c$ (rad/s) | Prediction | Simulation |
|---|---|---|---|---|
| 0 | 0 | 4.429 | oscillates | oscillates |
| 0 | 1.5 | 4.429 | oscillates | oscillates |
| 0 | 3 | 4.429 | oscillates | oscillates |
| $\pi/4$ | 0 | 2.397 | oscillates | oscillates |
| $\pi/4$ | 1.5 | 2.397 | oscillates | oscillates |
| $\pi/4$ | 3 | 2.397 | rotates | rotates |
| $\pi/2$ | 0 | 0.000 | rest | rest |
| $\pi/2$ | 1.5 | 0.000 | rotates | rotates |
| $\pi/2$ | 3 | 0.000 | rotates | rotates |

cases that start above the separatrix, ($\pi/4$, 3), ($\pi/2$, 1.5) and ($\pi/2$, 3). In these cases $P$ has enough energy to pass the top of the loop ($E_0 > mgR$), $\dot{\varphi}$ never becomes zero and $\varphi$ increases steadily, with $P$ going slowest at the top and fastest at the bottom. The speed alone is not enough: (0, 3) has the same $\dot{\varphi}_0$ as ($\pi/4$, 3) but less potential energy, and it oscillates. The period of the oscillations grows

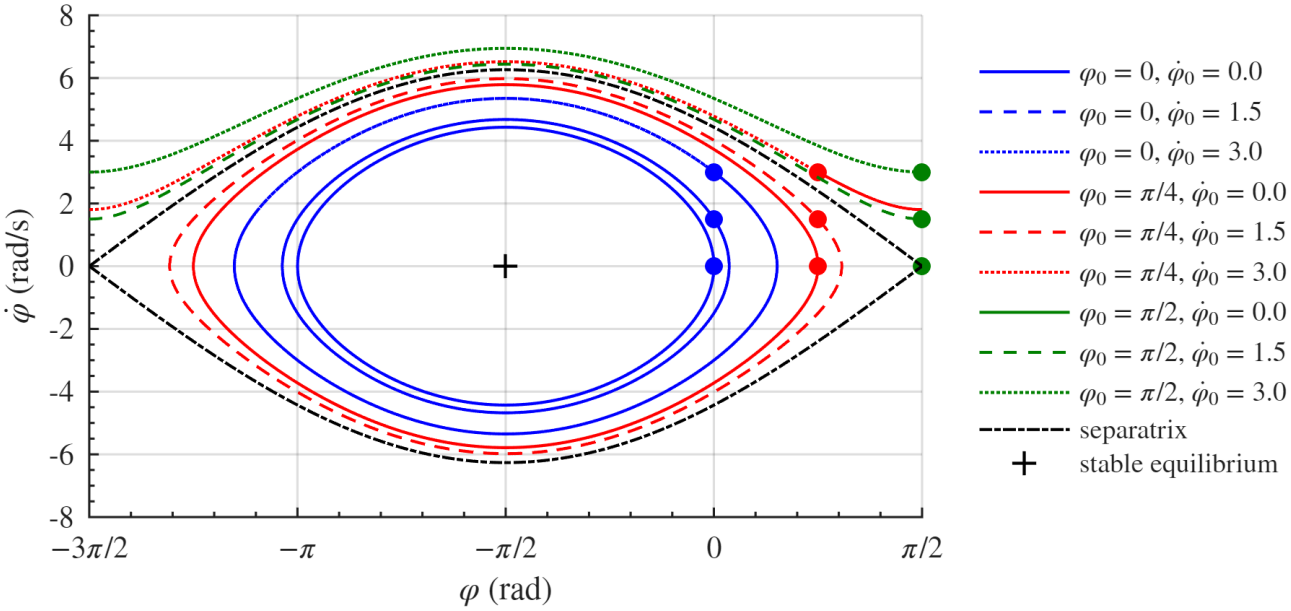



**Figure 6:** Phase portrait of the fixed loop ($A = 0$), with $\varphi_0$ in rad and $\dot{\varphi}_0$ in rad/s. Dots mark the initial conditions and the dash-dot line is the separatrix. Closed curves are oscillations and open curves are rotations.

from 2.37 s to 3.36 s as the curves get closer to the separatrix, because $P$ spends more time near the top, where it moves slowly. The case $(\pi/2,\, 0)$ starts at rest at the top of the loop, which is an unstable equilibrium position ($\ddot{\varphi} = 0$), so it stays at rest in the simulation, although any disturbance would make it fall.

### 4.6 Linearized Against Full Equation

Figure 7a shows the percentage error between the linearized and the full solution for the three values of $\omega$,

$$\text{error} = 100\,\frac{|\delta_{lin} - \delta_{full}|}{\max|\delta_{full}|}.$$

We divide by the maximum of $\delta_{full}$ and not by $\delta(t)$ because $\delta$ crosses zero. The error oscillates fast, so the thick lines show its envelope. Figure 7b shows $\delta(t)$ for $\omega = \omega_n/2$.

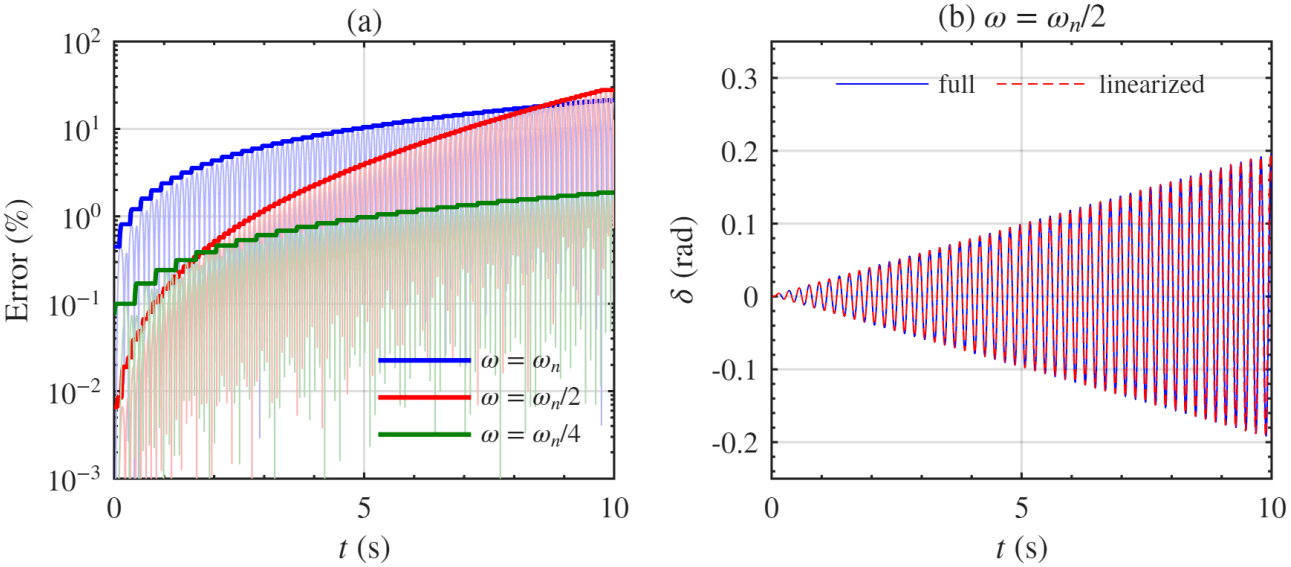



**Figure 7:** (a) Percentage error between the linearized and the full solutions (thin: error; thick: its envelope). (b) $\delta(t)$ of the full (solid) and linearized (dashed) solutions for $\omega = \omega_n/2$.

For $\omega = \omega_n/2$ there is resonance: the frequency of the centrifugal force, $2\omega$, is equal to the natural frequency, $\omega_n = \sqrt{g/R}$, so each push is in phase with the oscillation and the amplitude of $\delta$ grows linearly with time (Fig. 7b), up to 0.20 rad ($\approx 11°$) at 10 s. Then the hypothesis $\delta \ll 1$ is not true any more, $\sin\delta \approx \delta$ fails and the error grows up to 27.8%. In the full equation the pendulum is slower

when its amplitude is larger, so it gets out of phase with the forcing and the two solutions separate.

Outside resonance, the error is much smaller for $\omega = \omega_n/4$ (below 2%) than for $\omega = \omega_n$ (21.4%), although $\delta$ stays small in both cases (below 0.013 rad). In the linearization we neglected $\Omega^2\delta$ against $(g/R)\delta$ using the hypothesis $A^2 \ll g/(R\omega^2)$, that is $A^2\omega^2 \ll g/R$. Since $\Omega^2 \propto \omega^2$, both the forcing and the neglected term are 16 times larger at $\omega_n$ than at $\omega_n/4$, so the neglected term accumulates a larger phase difference between the two solutions as time passes.

# 5 Conclusions

- The particle has one degree of freedom, $\varphi$, and Eq. (2.13) contains only gravity and the centrifugal force. The Coriolis force is the largest inertia force, but it only loads the wire, through $N_2$. For $\omega = 0.1$ rad/s the rotation is slow, so $P$ swings as a pendulum of 90° amplitude and the wire carries up to $3mg$ in its plane.
- As we have derived in Eq. (2.17), $dE/dt = -\Omega x N_2$. This means that the mechanical energy is not conserved, because the moving wire does work, and it is conserved only for $A = 0$. $E$ changes in each swing but returns almost to its initial value when the plane stops.
- On a fixed loop, the energy gives the critical rate $\dot{\varphi}_c = \sqrt{(2g/R)(1 - \sin\varphi_0)}$, and all the simulated cases confirm it. Only three of the nine initial conditions rotate.
- We have seen that the linearized equation is a forced oscillator with a resonance at $\omega = \frac{1}{2}\sqrt{g/R}$, where $\delta \ll 1$ fails. Away from resonance it also loses accuracy with time, and the error is below 2% only for $\omega = \omega_n/4$.

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